

Trigonometry in 3D



1 All edges of the following cube are 7 cm long:

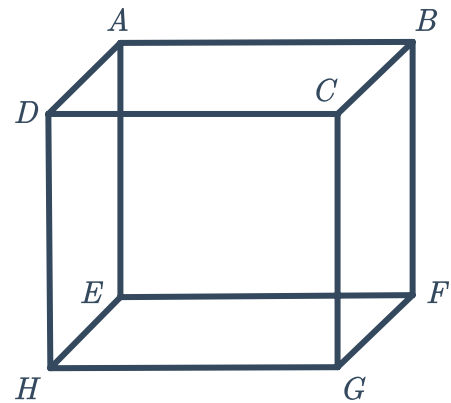
a Find the exact length of the following:

i EG ii AG

b Find the following angles to the nearest degree:

i $\angle EGH$ ii $\angle EGA$

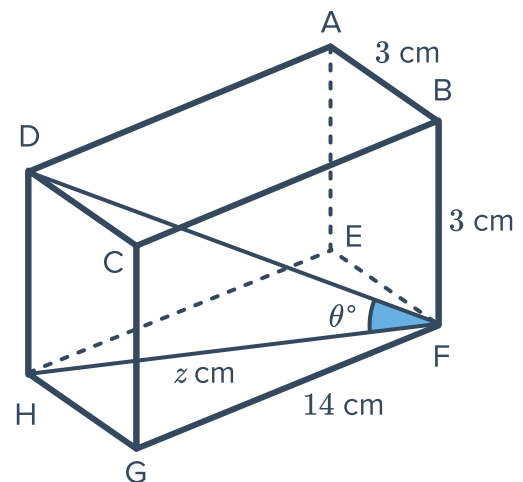
iii $\angle AHG$ iv $\angle AGH$



2 A square prism has sides of length 3 cm, 3 cm and 14 cm as shown in the diagram:

a If the diagonal HF has a length of z cm, find the value of z to two decimal places.

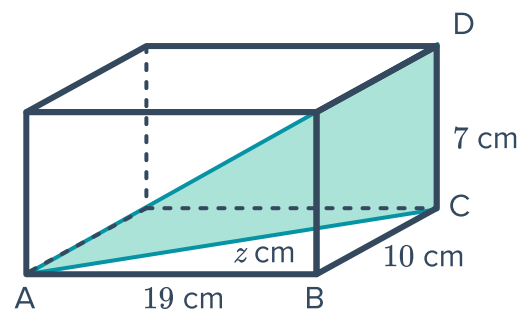
b If the size of $\angle DFH$ is θ° , find θ to two decimal places.



3 The box have a triangular divider placed inside it, as shown in the diagram:

a If $z = AC$, find the value of z to two decimal places.

b Find the area of the divider correct to two decimal places.





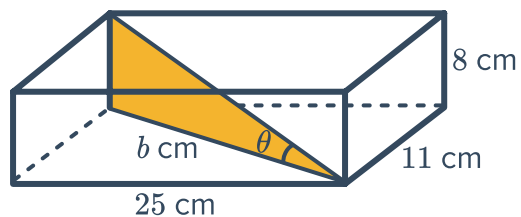
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A rectangular prism is shown with dimensions 12 cm (length), 9 cm (width), and 8 cm (height). A line segment of length y cm is drawn from the top-back-left corner to the bottom-front-right corner. A line segment of length x cm is drawn from the bottom-front-left corner to the bottom-back-right corner. The angle between these two segments is labeled θ .

A 3D diagram of a rectangular prism. The bottom-left edge is labeled 1 cm . The bottom-right edge is labeled $x\text{ cm}$. The back-right edge is labeled $y\text{ cm}$. The angle between the 1 cm edge and the diagonal of the bottom face is labeled α . The angle between the $x\text{ cm}$ edge and the diagonal of the right face is labeled β . The angle between the $y\text{ cm}$ edge and the diagonal of the left face is labeled γ . The angle between the two diagonals of the prism is labeled θ .

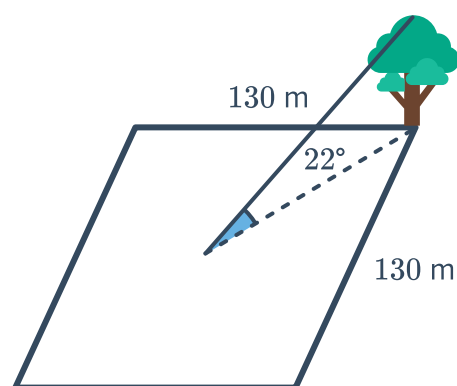
7 A $25\text{ cm} \times 11\text{ cm} \times 8\text{ cm}$ cardboard box contains an insert (the shaded area) made of foam:

- Find the exact length b of the base of the foam insert.
- Find the area of foam in the insert, rounded to the nearest square centimetre.
- Find the value of θ to the nearest degree.



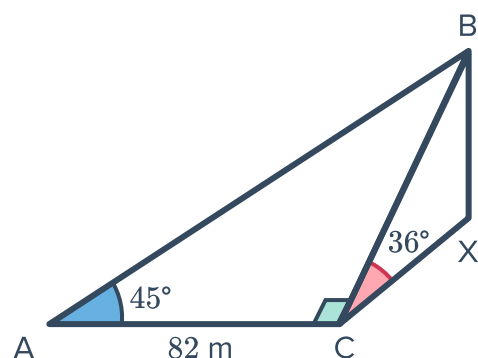
8 A tree stands at the corner of a square playing field. Each side of the square is 130 m long. At the centre of the field the tree subtends an angle of 22° as shown:

- Find the distance from the tree to the centre of the field. Round your answer to two decimal places.
- Find the height of the tree. Round your answer to two decimal places.
- Find the size of the angle subtended by the tree at the corner of the field opposite the tree. Round your answer to the nearest minute.
- Find the size of the angle subtended by the tree at an adjacent corner of the field. Round your answer to the nearest minute.

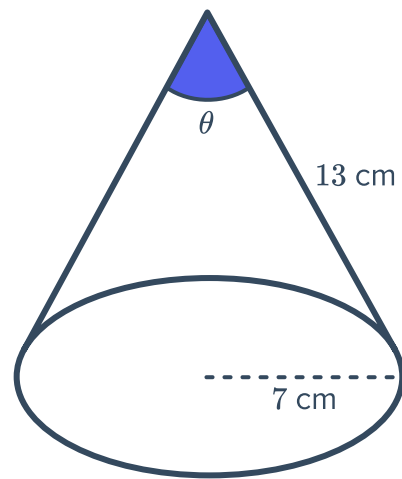


9 A , C , and X are three points in a horizontal plane and B is a point vertically above X as shown in the diagram:
Find the following lengths to the nearest metre:

- BC
- XB

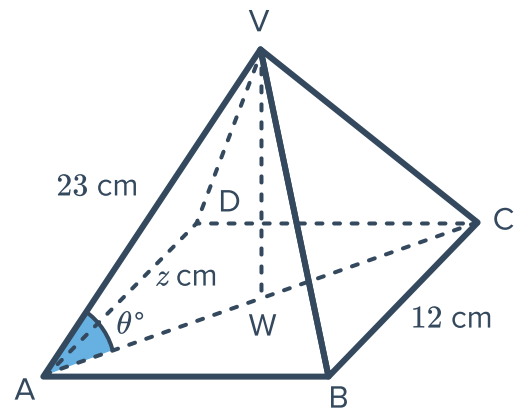


- 10 A cone has radius 7 cm and a slant height 13 cm:
Find the vertical angle, θ , at the top of the cone in degrees and minutes.



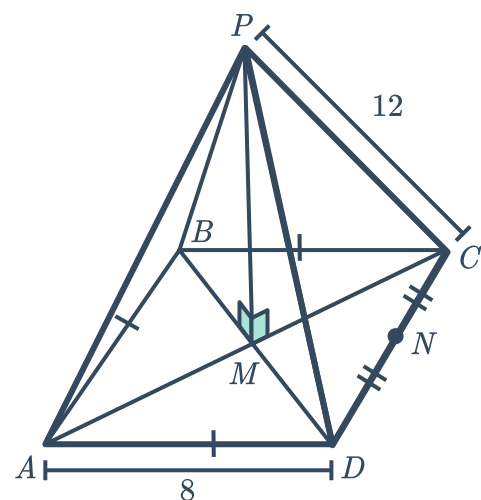
- 11 The following is a right pyramid on a square base with side length 12 cm. The edge length VA is 23 cm.

- If $z = AW$, calculate the length z .
- If $\theta = \angle VAW$, find the value of θ .

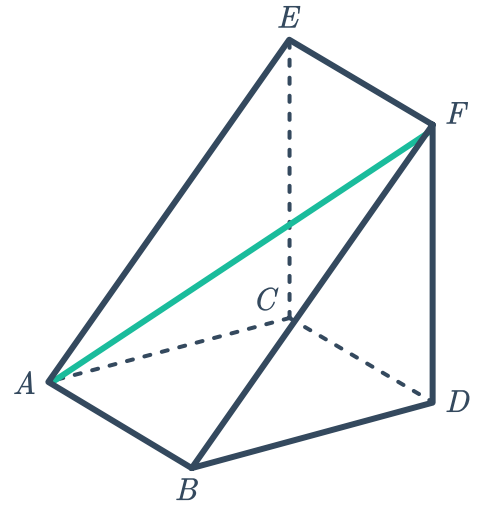


- 12 The following pyramid has a square base with side length 8 cm, and all slanted edges are 12 cm in length:

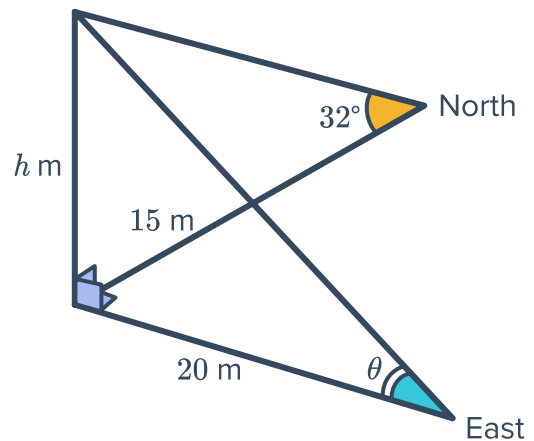
- Find the exact length of MD .
- Find the size of $\angle PDM$, correct to two decimal places.
- Find the exact height of the pyramid, MP .
- Find the length of MN .
- Find the exact length of PN .
- Find the size of $\angle PNM$, correct to two decimal places.
- Find the size of $\angle PDC$, correct to two decimal places.



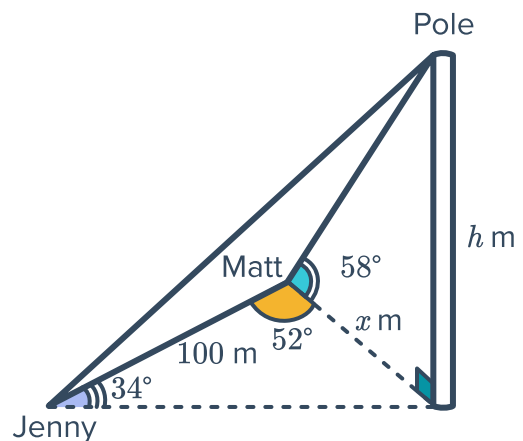
- 13 This triangular prism shaped box needs a diagonal support inserted as between A and F as shown:
 $AB = 19$, $BD = 30$ and $DF = 43$. find the length of AF to two decimal places.



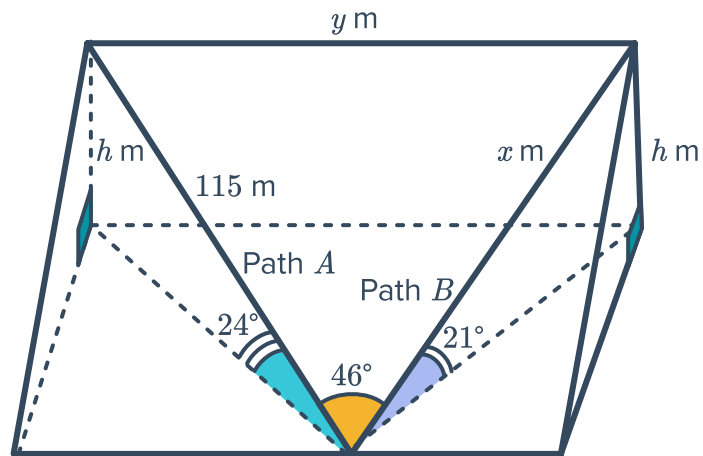
- 14 From a point 15 m due north of a tower, the angle of elevation of the tower is 32° .
- Find the height of the tower h . Round your answer to two decimal places.
 - Find the size θ of the angle of elevation of the tower at a point 20 m due east of the tower. Round your answer to the nearest degree.



- 15 A pole is seen by two people, Jenny and Matt:
- Matt is x m from the foot of the pole. Find x to the nearest metre.
 - Find the height of the pole h to the nearest metre.

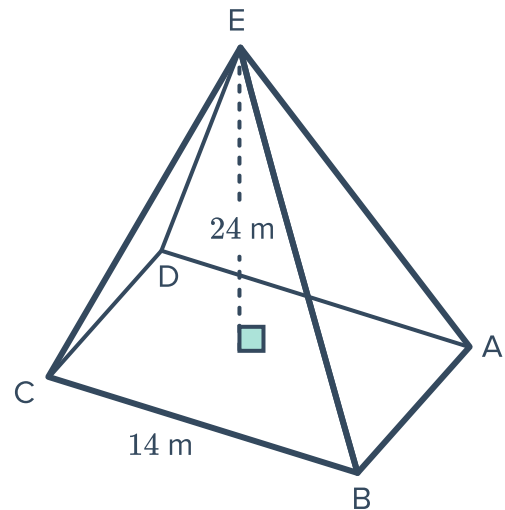


- 16 Two straight paths to the top of a cliff are inclined at angles of 24° and 21° to the horizontal:



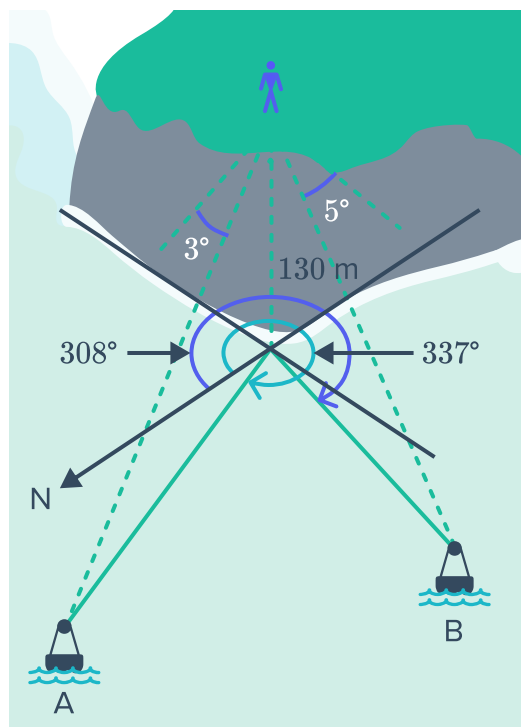
- If path A is 115 m long, find the height h of the cliff, rounded to the nearest metre.
 - Find the length x of path B , correct to the nearest metre.
 - Let the paths meet at 46° at the base of the cliff. Find their distance apart, y , at the top of the cliff, to the nearest metre.
- 17 A pyramid has a square base with side length 14 m and a vertical height of 24 m:

- Find the length of the edge AE . Round your answer to one decimal place.
- Find the size of $\angle BEA$, to the nearest minute.



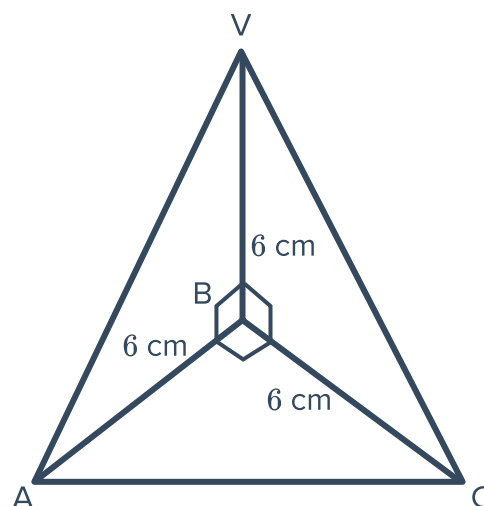
- 18 Two buoys, A and B , are observed from a lookout at the top of a 130 m high cliff. The bearing of buoy A is 337° and its angle of depression is 3° . The bearing of buoy B is 308° and its angle of depression is 5° :

- Find the distance, to the nearest metre, from buoy A to the base of the cliff.
- Find the distance from buoy B to the base of the cliff, to the nearest metre.
- Find the distance between the two buoys, to the nearest metre.



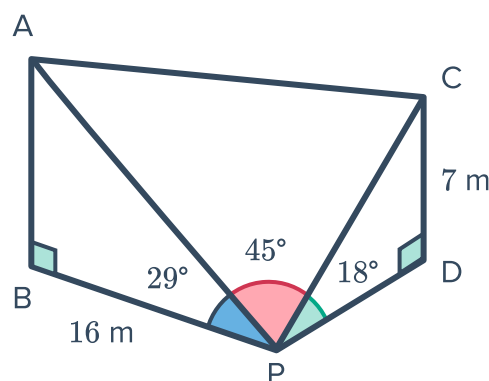
- 19 In the tetrahedron shown, the angles $\angle VBC$, $\angle VBA$, and $\angle ABC$ are all right-angles.

- Find the following distances, to two decimal places:
 - VA
 - VC
 - AC
- Find the size of angle $\angle VCA$. Round your answer to the nearest minute.



- 20 Roald is standing at point P and observes two poles, AB and CD , of different heights. P , B , and D are on horizontal ground: From P , the angles of elevation to the top of the poles at A and C are 29° and 18° respectively. Roald is 16 m from the base of pole AB . The height of pole CD is 7 m.

- Calculate the distance from Roald to the top of pole CD , to two decimal places.
- Calculate the distance from Roald to the top of pole AB , to two decimal places.
- Calculate the distance between the tops





- 21** Point A is due north of a tower, and has an angle of elevation to the top of the tower of 51° . Point B is 100 m from point A on a bearing of 120° . The angle of elevation from point B to the top of the tower is 25° .
Find the height of the tower to the nearest metre.
- 22** A room measures 5 m in length and 4 m in width. The angle of elevation from the bottom left corner to the top right corner of the room is 57° .
- a** Find the exact distance from one corner of the floor to the opposite corner of the floor.
 - b** Find the height of the room. Round your answer to two decimal places.
 - c** Find the angle of elevation from the bottom corner of the 5 m long wall to the opposite top corner of the wall. Round your answer to two decimal places.
 - d** Find the angle of depression from the top corner of the 4 m long wall to the opposite bottom corner of the wall. Round your answer to two decimal places.
- 23** A and B are two positions on level ground. From an advertising balloon at a vertical height of 740 m, A is observed in an easterly direction and B at a bearing of 159° . The angles of depression of A and B , as viewed from the balloon, are 30° and 20° , respectively.
- a** Find the distance from point A to the point directly below the advertising balloon, to the nearest metre.
 - b** Find the distance from point B to the point directly below the advertising balloon, to the nearest metre.
 - c** Find the distance between points A and B , to the nearest metre.